

PHC: Public Health Concentration Courses

Courses

PHC 2082 Informatics and Your Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Multi-disciplinary exploration of how information is represented, processed, shared, and protected in tools/applications directly linked to the health of you and the planet. Examines impacts on individuals and public health; provides practice with a variety of digital technologies and data collection strategies; addresses interpreting results of and concerns in medical and health-related human subject research. Develops professional and technical skills, including presenting ideas through written/verbal communication, citing AI tools as part of AI-facilitated learning, and applying ethical protocols within AI-augmented workflows for visualizing data. Students will use the scientific method to formulate hypotheses in map and data dashboard design processes to create visualizations to explain observed patterns and statistical models derived from their collected geolocated data to foster an understanding of scientific principles and their practical application toward real-world issues. Meets General Education requirement in Natural Sciences.

PHC 3767 Globalization and Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course examines the role of globalization in the burden of disease and global cooperation in development of interventions that promote health and avert transmission of transboundary pathogens across national boundaries and continents. The course will discuss the role of multinational and bilateral global collaborations for addressing health issues that cross national borders. The emerging threats require consistent multinational approaches and an informed public health workforce at all levels of practice. The course will use case studies to examine the impact of globalization on human health.

PHC 4024 Applied Research Methods in Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

In this applied course, students will learn how to conduct and interpret research in population health and disease, building on their knowledge from epidemiology and biostatistics and other content areas in public health. The course will provide skills in critically reading the literature related to public health-related research, developing a research question and a testable hypothesis, creating an analysis plan, and dissemination of results in paper and digital modes. The course will cover quantitative and qualitative research methods commonly used in public health research and practice.

PHC 4048 Epidemiology for Public Health Practice

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course delves into the practical application of epidemiological methods and principles to address real-world public health challenges. Students will engage with hands-on activities and case studies to understand how epidemiology informs and supports public health practice and decision-making. The course covers the application of epidemiological techniques to analyze health data and trends, as well as the design, implementation, and evaluation of public health interventions based on epidemiological evidence. Methods for assessing and responding to disease outbreaks, health emergencies, and chronic disease management will be addressed, alongside the use of epidemiological data to develop public health policies and programs. Techniques for effectively communicating epidemiological findings to stakeholders and the public will be developed within projects that involve real data sets and case scenarios that emphasize best practices for translating epidemiological research into actionable strategies for improving community health outcomes.

PHC 4101 Essentials of Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course provides an overview of the history and philosophy of public health, as well as its core values, concepts, and functions across the globe and in society. The course also covers basic principles and tools of population health, as well as the factors leading to disease causation and disease prevention.

PHC 4109 Diseases in Human Populations

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides students with an overview of the biological basis of public health and the implications of major communicable and non-communicable diseases of public health significance at the state, national, and international levels. Diseases contributing to national and global health disparities and strategies for reducing disease burden are discussed. Before enrolling, students must have at least one semester of a college science such as biology or a comparable course.

PHC 4140 Public Health Planning and Analysis

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides students with an overview of public health planning with an emphasis on the use of public health principles and theories of program planning, implementation, and evaluation. Students will gain an understanding of the complex factors that impact needs assessment. Case studies will be analyzed using examples of national, state, or local health programs and interventions.

PHC 4320 Environmental and Occupational Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course is designed to provide undergraduate public health students with essential knowledge and applied skills in environmental health, with a focus on biologic, chemical, physical, and radiation hazards that pose risks to communities and the workforce. By integrating scientific, regulatory, and interprofessional approaches, the course strengthens community and workforce readiness to prevent, detect, and respond to environmental and occupational threats impacting human health and safety.

PHC 4464 Understanding Health Equity and Health Disparities

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course introduces students to social determinants of health, health equity, and health disparities in the United States. This course provides students with an overview of the societal impacts of racism, poverty, and inequity and the ways these constructs are contributing health disparities among individuals and communities. Students will explore sources of data for health disparities analysis. The course further explores the role of major national stakeholders and programs addressing social determinants of health contributing to disparities and implementing evidence-based strategies for eliminating health inequalities.

PHC 4533 Foundations of Maternal and Child Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Maternal and Child Health (MCH) is an interdisciplinary field that focuses on understanding the determinants, mechanisms, and systems that ensure the health, safety, well-being, and optimal development of mothers, children, and families across diverse communities. This course addresses critical global health challenges such as maternal and infant mortality, malnutrition, and infectious diseases by exploring population-level interventions, health policies, and programs. Through a combination of weekly lectures, case studies, and hands-on projects, students will engage with the history of MCH, current research, key programs and policies, and emerging global issues. The course is designed to meet the educational needs of both undergraduate and graduate students, with additional assignments and expectations for graduate students to ensure a deeper understanding and application of course concepts. Offered concurrently with PHC 5530. Graduate students will be assigned additional work.

PHC 4666 Pandemics and Microbial Threats to Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course addresses the causes and effects of pandemics and their effect on human populations, including economic and social disruptions from emerging and re-emerging microbial threats. Through the examination of both historical and contemporary pandemics, students will gain insights into their evolution and impact, while fostering an appreciation for the challenges faced by global health systems. The course emphasizes an interdisciplinary approach, integrating principles from epidemiology, microbiology, and public health to address these complex issues. By equipping students with the tools and knowledge necessary for effective public health practice, this course prepares future professionals to contribute to global health security and response efforts.

PHC 4712 Applied Qualitative Research Methods

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides undergraduate students with essential skills in qualitative research methods within the context of public health. It begins with an overview of the goals, purposes, ethics, and logic of social research, comparing quantitative and qualitative approaches. The focus then shifts to hands-on learning, where students will practice qualitative data collection techniques such as field observation, qualitative interviewing, and writing fieldnotes. The course also covers strategies for analyzing qualitative data and interpreting findings in public health. By the end of the course, students will be equipped to confidently design and conduct their own qualitative research projects, applying these methods to address complex public health issues. Offered concurrently with PHC 5198; graduate students will be assigned additional work.

PHC 4905 Directed Study

College of Health, Department of Public Health

1-12 sh (may be repeated indefinitely for credit)

PHC 4930 Public Health Seminar

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This public health seminar course presents current topics in the public health field at the local, regional, national, and international levels that may not be taught in traditional public health courses. Students employ critical thinking in analyzing and articulating opposing viewpoints on contemporary public health problem topics.

PHC 4940 Public Health Capstone

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides students with opportunities for practical and experiential high impact practices. The course serves a capstone requirement for undergraduate Public Health majors and is designed to allow students to review, integrate, and apply concepts and methods presented in the foundational core courses. All students complete a cumulative, integrative, and scholarly, and applied experience or inquiry project that serves as a capstone to the BSPH educational experience. Students develop an ePortfolio documenting their experiential learning. The capstone requirement must be taken in the senior year.

PHC 5050 Biostatistics for Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This is a second course in biostatistics for students in the graduate Public Health program. The topics include descriptive statistics, probability, standard probability distributions, sampling distributions, point and confidence interval estimation, hypothesis testing, power and same size estimation, one and two-sample parametric and non-parametric methods for analyzing continuous or discrete data, simple linear regression, logistic regression and other multivariate methods. Students will use a statistical software package for data management and statistical analyses. This is a fully online course with its own office hours and discussions. STA 2023 or equivalent is a pre-requisite for this course (see UWF Catalog). It is important to have a good understanding of inferential statistics, such as confidence intervals and test of hypotheses (for two samples). Students must have completed STA 2023 or equivalent in college.

PHC 5102 Principles of Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course introduces students to the history of public health, the structure of the public health system and the various sectors of public health practice, in order to gain an understanding of the complex factors that determine the health status of a community. Lectures will draw from the public health field and also related disciplines such as behavioral sciences, healthcare management, medical ethics, and social work. National, state, and local level practices will be analyzed, as well as the role that law and government play in the public's health. This course meets the foundational public health knowledge area learning outcomes defined by the Council on Education for Public Health. Students admitted into the MPH Program, regardless of concentration, must complete this foundational course within their first two semesters of enrollment.

PHC 5108 Monitoring and Evaluation in Global Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course provides students with fundamental frameworks for monitoring and evaluation in global health practice. The course provides an overview of different types of program evaluation, including needs assessment, formative research, process evaluation, monitoring of outputs and outcomes, and impact assessment. Students gain practical experience through a series of case-studies involving the use of a conceptual framework in the design of an evaluation plan, that includes development of indicators, analysis of computerized service statistics, and impact measurement. The course addresses experimental, quasi-experimental, and non-experimental study designs, including the strengths and limitations of these designs in population and global health practice.

PHC 5123 Biological Basis of Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

An overview of scientific principles of public health and their application to public health problems with significant state, national and international impact. It is recommended that students have at least one semester of a college science such as, biology or a comparable course before enrolling.

PHC 5198 Applied Qualitative Research Methods

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides graduate students with essential skills in qualitative research methods within the context of public health. It begins with an overview of the goals, purposes, ethics, and logic of social research, comparing quantitative and qualitative approaches. The focus then shifts to hands-on learning, where students will practice qualitative data collection techniques such as field observation, qualitative interviewing, and writing fieldnotes. The course also covers strategies for analyzing qualitative data and interpreting findings in public health. By the end of the course, students will be equipped to confidently design and conduct their own qualitative research projects, applying these methods to address complex public health issues. Offered concurrently with PHC 4712; graduate students will be assigned additional work.

PHC 5355 Fundamentals of Occupational Safety and Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course provides an overview of occupational safety and health and is a foundation for the advanced study of hazards in work settings. Students are provided an introduction to safety and the health fields, an overview of loss control information, and an analysis relevant to OSHA standards. Industry-specific principles for the development and management of occupational safety and health programs, resolution of safety and health issues, and improved safety performance are presented.

PHC 5356 Fundamentals of Industrial Hygiene

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course provides a foundation in the multidisciplinary approach to the study of industrial hygiene intended for a wide range of health-related professionals who serve in critical roles for recognizing, evaluating, and controlling environmental or occupational hazards. The course provides students with insight into the management of occupational health hazards and diseases that can be leveraged in professional practice.

PHC 5410 Social and Behavioral Sciences in Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Covers behavioral and social science contributions to science disciplines, including psychology, sociology, and anthropology, will be reviewed and integrated with public health objectives and outcomes. Using a biopsychosocial framework, the role of social, psychological, and behavioral factors in health and illness are emphasized.

PHC 5442 Global Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course will introduce students to the main concepts of the public health field and the critical links between global health and social and economic development. Students will get an overview of the determinants of health, and how health status is measured. Students will also review the burden of disease, risk factors, and key measures to address the burden of disease in cost-effective ways. The course will review specific topics related to the most important communicable and non-communicable diseases as well as issues related to food distribution, reproductive health and other global major health concerns with an important focus on low- and middle-income countries and on the health of the poor. We will also discuss cross-cutting global health issues such as poverty and equity, human rights and ethical issues in public health; globalization and health and complex emergencies.

PHC 5530 Foundations of Maternal and Child Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Maternal and Child Health (MCH) is an interdisciplinary field that focuses on understanding the determinants, mechanisms, and systems that ensure the health, safety, well-being, and optimal development of mothers, children, and families across diverse communities. This course addresses critical global health challenges such as maternal and infant mortality, malnutrition, and infectious diseases by exploring population-level interventions, health policies, and programs. Through a combination of weekly lectures, case studies, and hands-on projects, students will engage with the history of MCH, current research, key programs and policies, and emerging global issues. The course is designed to meet the educational needs of both undergraduate and graduate students, with additional assignments and expectations for graduate students to ensure a deeper understanding and application of course concepts. Offered concurrently with PHC 4533. Graduate students will be assigned additional work.

PHC 5905 Directed Study

College of Health, Department of Public Health

1-12 sh (may be repeated indefinitely for credit)

PHC 6000 Epidemiology for Public Health Professionals

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This foundational course covers the application of epidemiologic procedures to the understanding of the occurrence and control of conditions such as infections and chronic diseases, mental disorders, community, and environmental health hazards, accidents, and geriatric problems in human populations. The course is critical to developing student competency in the foundational and practical utilities of epidemiology as a tool for disease surveillance and outbreak investigations, disease prevention and treatment. Part of the Master of Public Health degree program.

PHC 6003 Chronic Diseases Epidemiology

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course is designed to give the student current and comprehensive information on the epidemiology, etiology, pathogenesis, risk factors and preventive measures of common chronic diseases at the population level. The course will cover selected topics in chronic disease with an emphasis on disease occurrence in the United States, Florida and the current status of local research projects.

PHC 6005 Urbanization and Population Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course focuses on the impact of urbanization and the transmission of disease-causing organisms, as well as the interaction between human behavior and environmental changes on population health. Factors such as overcrowding, access to quality housing, modernized urban amenities, lifestyle choices and sanitation(WASH) contribute to the spread of disease in urban areas in developed and low-to-middle income countries.

PHC 6015 Epidemiological Research Designs and Methods

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course covers research design and methods commonly used in epidemiology and public health research. The course covers both quantitative and qualitative research designs, including, observational, quasi-experimental, and experimental designs used in epidemiological investigations. Methods for reliable and valid data collection and analysis will be covered. Common statistical methods for the analysis of public health data are discussed.

PHC 6110 Comparative Health Systems

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course will cover the major models for provision and financing of healthcare used around the world today. The historical development of these models will be traced, and the societal values and other factors underlying countries' choice of healthcare systems and policies will be examined. Students will learn to analyze the advantages and disadvantages of various ways of organizing and financing healthcare and to evaluate health policies according to a range of criteria for cost, quality, and equity.

PHC 6150 Public Health Policy

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The course explores general principles of planning, management, and evaluation of health care programs, policies and interventions implemented by public and private organizations. The goal of the course is to ensure a broad understanding of critical issues concerning the organization, delivery, and financing of public health and health care in the United States and examines the role of private, community-based organizations, federal, state, and local governments in ensuring healthy communities. The basic conceptual frameworks underlying healthcare decision making and assessment of the financing, organization, outcomes and delivery of healthcare services are presented.

PHC 6194 GIS Applications in Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

An online course providing hands-on training in the use of geographic information systems for public health-related data. Students will complete projects covering the collection, analysis, and visualization of spatial data using both public domain and commercial software tools supporting geospatial data. Through a set of focused case studies, students will learn the basic features and limits of each tool, as well as interoperability with other GIS software products (both public domain and commercial packages). Part of the Master of Public Health degree program.

PHC 6196 Applied Data Analysis in Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides an applied, hands-on experience in public health data analysis and scientific writing. Students will develop a research hypothesis, manage and analyze real-world public health datasets using statistical software, interpret results, and produce a complete scientific manuscript. Topics include data collection, data management, statistical analysis, interpretation, and effective reporting of results. Enrollment assumes prior coursework or training in epidemiology and statistics, as well as familiarity with statistical reasoning. Students are expected to have prior experience using at least one statistical software package (e.g., SPSS, Stata, SAS, R, Python, or equivalent). Completion of PHC 6000, PHC 6300, PHC 5410, and PHC 5050 (or equivalents) is strongly recommended before enrolling.

PHC 6251 Disease Surveillance and Monitoring

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Disease surveillance and monitoring is the systematic collection, analysis, interpretation, and dissemination of data for use in prioritizing, planning, implementing, and evaluating health programs, activities and practices in the United States as well as in other developed and developing countries. We will focus on these fundamental processes and procedures which are utilized to investigate and track infectious and communicable diseases as well as non-infectious chronic diseases. The course will highlight the importance of designing and reporting quantitative and qualitative contents in disease surveillance.

PHC 6300 Environmental Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This foundational public health course provides a broad overview of the relationship between humans and their environment, and of the efforts to prevent or mitigate environmental threats. Students will be given an overview of the chemical, physical, and biological hazards present in our living and working environment and their effects on human health. The course is designed to acquaint students with the scientific and technical foundations of population health and examines both the practice of environmental health and problems addressed public health practitioners.

PHC 6310 Environmental Toxicology

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Environmental toxicology is the study of the effects of toxic substances on health and the environment. The student will recognize that human survival depends upon the well-being of other species and upon the availability of clean air, water, and food; and anthropogenic, as well as naturally occurring, chemicals can have detrimental effects on living organisms and ecological processes. Concepts to be covered include occurrence of toxicants, damage process and action of toxicants, factors affecting xenobiotic action, defense responses to toxicants, and others. Will also examine chemicals of environmental interest and how they are tested and regulated. Case studies and special topics will be examined.

PHC 6347 Aerospace and Occupational Toxicology

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course is an overview of practical hazards, assessment, and mitigation techniques used in the unique area of aerospace. The course provides an overview of the chemical, physical, and biological hazards present in our living and working environment and their effects on human health. The course is an elective for students in the MPH program who are participating in the Aerospace Medicine residency program and open to other students with the permission of the Department of Public Health.

PHC 6360 Accident Investigation and Risk Management

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

Accident Investigation and Risk Management includes an aerospace safety overview, biomechanics of impact, restraint systems, crew protection, and crew escape concepts, aviation and space vehicle crashworthiness, aerospace injury mechanisms, conduct of an accident investigation, forensic concepts, legal issues, and promoting prevention strategies to avoid future accidents. This course is an elective restricted to students in the MPH degree program who are participating in the aerospace and occupational medicine residency program.

PHC 6585 Communication in Public Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course explores the critical role of communication in public health, focusing on strategies and practices essential for effective public health messaging and advocacy. Students will develop skills in crafting, delivering, and evaluating health communication campaigns aimed at diverse audiences, including the general public, policymakers, and healthcare professionals. Through case studies and practical exercises, students will gain hands-on experience in public health communication, preparing them to address real-world challenges and improve public health outcomes through effective messaging.

PHC 6670 Ethical Issues in Global Health

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The purpose of this course is to provide an overview of major ethical principles and theories applied to global health, and how culture influences ethical decision-making in health resource allocation. Students will examine some of the primary theories and principles in global health practice and research ethics including virtue, deontology, utilitarian, autonomy, justice, beneficence, and nonmaleficence. The course explores prominent global health issues to build greater knowledge and understanding of global health ethics as a vital tenant of effective decision-making to promote optimal health. Topics will address broad ethical issues related to pandemic preparedness, ethical human subject research in developing countries, human rights, gender based violence, resource allocation, and the effects of globalization on health and quality of life within countries.

PHC 6676 Public Health Response in Humanitarian Emergencies

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This course provides an overview of the types of humanitarian emergencies and their public health significance. Students explore the public health impact of emergencies and disasters in low-to-middle income countries. Students are introduced to population health needs assessment during emergencies and key indicators needed for timely response and mitigation. Emphasis is placed on the coordination among local, regional, national, governmental, and non-governmental organizations in the delivery of public health services.

PHC 6905 Directed Study

College of Health, Department of Public Health

1-12 sh (may be repeated indefinitely for credit)

PHC 6945 Applied Public Health Practice Experience

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

This Applied Public Health Practice Experience provides students with an applied, scholarly, and mutually beneficial experience in a public health practice setting. It is completed under the supervision of an approved preceptor and a faculty member teaching in the UWF MPH program. The student will work on a problem related to management, development, or administration of a program in public health or related to research in public health. This experience must reinforce academic course work, meet community or agency needs, and provide the student with an opportunity to integrate and apply relevant MPH program competencies consistent with current accreditation standards. All students must produce a minimum of two practice artifacts approved by the Department of Public Health, as well as an ePortfolio showcasing these deliverables. The course is graded on a Satisfactory / Unsatisfactory scale. Permission is required. Students are expected to complete MPH foundational coursework prior to enrollment.

PHC 6946 Public Health Integrative Learning Experience

College of Health, Department of Public Health

3 sh (may not be repeated for credit)

The Public Health Integrative Learning Experience course is the culminating experience of the MPH Program that demonstrates synthesis of foundational and concentration competencies consistent with current accreditation standards. It is completed under the supervision of a faculty member teaching in the UWF MPH program. The student produces a high-quality written product that is appropriate for the student's educational and professional objectives and approved by the Department of Public Health. Written products might include the following: research study design and implementation, program evaluation report, training manual, policy statement, legislative testimony with accompanying supporting research, etc. The written product is developed and presented orally to program faculty and relevant stakeholders. The course is graded on a Satisfactory / Unsatisfactory scale. Permission is required. Students are expected to complete MPH foundational coursework prior to enrollment and register for this course at or near their final semester in the program.